SYED MAOZ ARIF

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SUMMARY

Detail-oriented Hardware Developer and Electronics Engineer with a proven track record in multi-layer PCB design, embedded systems, and IoT architecture. Developed 12+ real-world engineering projects and achieved 17 technical certifications. Recognized for innovative problem-solving, securing 1st Position in national engineering exhibitions and a Global Instructables Contest win. Adept at cross-functional leadership, managing 50+ members as IET Chapter President, and delivering high-quality freelance engineering solutions with a 100% client satisfaction rate.

EDUCATION

COMSATS University Islamabad, Abbottabad Campus — Expected 2028
B.S. Electrical Engineering (Electronics) | GPA: 3.61/4.0 — *Abbottabad, Pakistan*

- **Honors:** Altium Global Scholarship Awardee (Selected among top 1% globally for engineering excellence).
- **Relevant Coursework:** Embedded Systems, Digital Logic Design, Signal Integrity, C/C++ Programming.

WORK EXPERIENCE

Freelance Hardware & Mechanical Designer — Jan 2024 – Present
Fiverr & Upwork — *Remote*

- Delivered 15+ multi-layer PCB designs and 3D CAD models to international clients, maintaining 100% on-time delivery and 5-star satisfaction across 4 industry sectors.
- Optimized component selection (BOM) and schematic captures, reducing overall hardware prototyping costs by 25% for 10+ clients.

President (Previously VP & Membership Head) — May 2025 – Present
IET On Campus, CUI Abbottabad — *Abbottabad, Pakistan*

- Directed chapter operations and managed a 5-person cross-functional executive team, fostering engineering excellence for 50+ active members.
- Organized 5+ technical workshops (e.g., AI in UAVs) with 100+ attendees, increasing year-over-year student engagement by 40%.

PROJECTS

Industrial Edge AI Condition Monitoring & 3D Digital Twin — Jul 2026

- Designed a dual-MCU industrial development board integrating 70 GPIOs, reducing 3-phase induction motor diagnostic time by 40% for predictive maintenance applications.
- Secured 1st Position (Software Category) at CEP Exhibition SP26 and 2nd Position at EMCOT 2026 out of 50+ competing projects.

Stratum-1 Precision Hardware Clock & GPS Bridge — Aug 2026

- Engineered a hardware time synchronizer using pure digital logic, bypassing OS latency to achieve 0ms delay and 100% synchronization accuracy.
- Developed a custom Verilog UART receiver and NMEA state machine on a Lattice FPGA to process live GPS signals directly in silicon.

ElectroParts IMS: Inventory Architecture — Aug 2026

- Architected a cross-platform inventory management system supporting 32 distinct electronics categories, increasing component retrieval efficiency by 50%.
- Implemented dual-storage local/cloud synchronization and linear barcode generation, enabling zero-latency data access for 1000+ logged components.

IncliSense Precision Digital Level — Apr 2026

- Built a precision leveling device analyzing dual-axis spatial data via I2C, yielding high-accuracy PC telemetry for real-time monitoring.
- Awarded Global Runner-Up in the Instructables Sensors Contest out of 500+ international submissions.

TECHNICAL SKILLS

Hardware Design: Altium Designer, KiCad, Proteus, LTspice, Multisim, Schematic Capture, Signal Integrity
Embedded Systems: ESP32, STM32, Arduino, Lattice FPGA, I2C, SPI, UART, MQTT, Firebase
Programming: C/C++, Python, Verilog (HDL), MATLAB, JavaScript, HTML5/CSS3
3D CAD & Tools: SolidWorks, Fusion 360, AutoCAD, 3D Printing, Git, LaTeX, Agile Methodologies